

ASSOCIATION ANALYSIS

EVALUATION – PART I

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EVALUATION

The following concepts will be introduced:

- ✓ **INTERESTING/UNINTERESTING ASSOCIATION RULE**
- ✓ **LIFT**
- ✓ **INTEREST FACTOR**
- ✓ **CORRELATION ANALYSIS**
- ✓ **IS MEASURE**





Once you have generated a **set of patterns** which are potentially useful for the Director of the mini-market, you must **rank them in order of interestingness**. What is trash for you could be gold for the Director of the mini-market or vice versa.

It is important to establish a set of well accepted **criteria for evaluating the quality of association patterns**.

Two sets of criteria:

➤ **STATISTICAL ARGUMENTS**

➤ **SUBJECTIVE ARGUMENTS**



- Patterns involving a set of mutually independent items
- Patterns covering very few transactions



UNINTERESTING

(may capture spurious relationships in the data)



They can be eliminated by applying an

OBJECTIVE INTERESTINGNESS MEASURE

USES STATISTICS DERIVED FROM DATA TO DETERMINE
WHETHER A PATTERN IS INTERESTING

SUPPORT, CONFIDENCE, CORRELATION

Two sets of criteria:

➤ **STATISTICAL ARGUMENTS**

➤ **SUBJECTIVE ARGUMENTS**



A pattern is considered **SUBJECTIVELY UNINTERESTING** unless it reveals unexpected information about the data or provides useful knowledge that can lead to profitable actions.



UNINTERESTING

(despite having high support and confidence, because the relationship represented by the rule may seem rather obvious)



INTERESTING

(unexpected and may suggest cross-selling opportunity for retailers)

Two sets of criteria:

➤ STATISTICAL ARGUMENTS

➤ **SUBJECTIVE ARGUMENTS**

Incorporating subjective knowledge into pattern evaluation is a difficult task because it requires a considerable amount of prior information from the domain experts



An **OBJECTIVE INTERESTINGNESS MEASURE** is

- a **data-driven** approach for evaluating the quality of association patterns
- **domain-independent** and requires minimal input from the users (threshold only)
- **computed based on** the frequency counts tabulated in a **CONTINGENCY TABLE**

	B	not B	
A	f_{11}	f_{10}	f_{1+}
not A	f_{01}	f_{00}	f_{0+}
	f_{+1}	f_{+0}	N

f_{1+} = support count for A

f_{+1} = support count for B

A = item A is present in the transaction

not A = item A is not present in the transaction

f_{10} = # of transactions where item A is present (**A**) while item B is not present (**not B**)



Suppose the Director of the mini-market is interested in analyzing the relationship between people who drink tea and coffee. The following contingency table is obtained from the available transactions.

	coffee	not coffee	
tea	150	50	200
not tea	650	150	800
	800	200	1,000

The information given in this table can be used to evaluate the following association rule



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support = 15%

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	coffee	not coffee	
tea	150	50	200
not tea	650	150	800
	800	200	1,000

confidence = 75%

support = 15%



The information given in this table can be used to evaluate the following association rule



At a first glance, it may appear that people who drink tea also tend to drink coffee



Suppose the Director of the mini-market is interested in analyzing the relationship between people who drink tea and coffee. The following contingency table is obtained from the available transactions.

	coffee	not coffee	
tea	150	50	200
not tea	650	150	800
	800	200	1,000

The fraction of **people who drink coffee**, regardless of whether they drink tea, is **80%**, while the fraction of **tea drinkers who drink coffee** is only **75%**.



The information given in this table can be used to evaluate the following association rule



At a first glance, it may appear that people who drink tea also tend to drink coffee



Suppose the Director of the mini-market is interested in analyzing the relationship between people who drink tea and coffee. The following contingency table is obtained from the available transactions.

	coffee	not coffee	
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	800	200	1,000

Ignores the support of
the itemset in the rule
consequent



Misleading despite its
high confidence value

The fraction of people who drink coffee, regardless of whether they drink tea, is 80%, while the fraction of tea drinkers who drink coffee is only 75%.



The information given in this table can be used to evaluate the following association rule



At a first glance, it may appear that people who drink tea also tend to drink coffee

support = 15%

confidence = 75%



INTEREST FACTOR: addresses the problem of confidence, i.e., ignoring the support of the itemset appearing in the rule consequent, we start from the **LIFT**

$$Lift = \frac{c(A \rightarrow B)}{s(B)}$$



LIFT is the ratio of the rule confidence to the support of the consequent.

For binary attributes the **LIFT** is equivalent to another objective measure called

INTEREST FACTOR, which is defined as follows:

$$I(A, B) = \frac{s(A, B)}{s(A)s(B)} = \frac{Nf_{11}}{f_{1+}f_{+1}}$$

= 1 if A and B are independent

> 1 if A and B are positively associated

< 1 if A and B are negatively associated

TEXT MINING

TO EVALUATE ASSOCIATION BETWEEN PAIR OF WORDS

	B	not B	
A	f_{11}	f_{10}	f_{1+}
not A	f_{01}	f_{00}	f_{0+}
	f_{+1}	f_{+0}	N

Compares the frequency of a pattern against a baseline frequency computed under the *statistical independence assumption*.



CORRELATION ANALYSIS: analyzes relationships between a pair of attributes. Continuous attributes are analyzed using Pearson's correlation coefficient. Correlation for binary attributes is measured using the ϕ -coefficient

$$\phi = \frac{f_{11}f_{00} - f_{01}f_{10}}{\sqrt{f_{1+}f_{+1}f_{0+}f_{+0}}}$$

which ranges from -1 to +1. If the items are statistically independent then it's value is zero, i.e., $\phi = 0$.

It is more suitable for analyzing **symmetric binary attributes**.



IS MEASURE: allows to handle asymmetric binary attributes

$$IS(A, B) = \sqrt{I(A, B)s(A, B)} = \frac{s(A, B)}{\sqrt{s(A)s(B)}}$$

It is large when the interest factor and support of the pattern are large.

In the case where **A and B are independent** we have

$$IS(A, B) = \frac{s(A, B)}{\sqrt{s(A)s(B)}} = \frac{s(A)s(B)}{\sqrt{s(A)s(B)}} = \sqrt{s(A)s(B)}$$

It shares a similar problem as the confidence measure, the value of the measure can be quite large, even for uncorrelated and negatively correlated patterns.



TEXT MINING
TO EVALUATE ASSOCIATION
BETWEEN PAIR OF WORDS

